

VizTrust: A Visual Analytics Tool for Capturing User Trust Dynamics in Human-AI Communication

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Project page 



xin-wang-kr.github.io/VizTrust-CHI25/

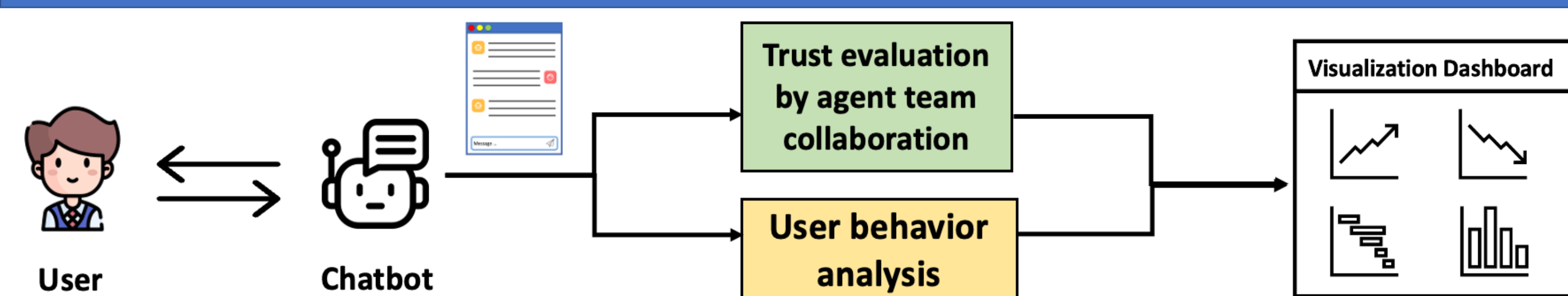
Challenge to Understand User Trust

- **Operationalization challenge:** User trust is **dynamic, contextual, and multi-dimensional**, making it resistant to precise quantification.
- **Method challenge:** Traditional approaches, such as surveys, fail to capture the temporal nature of trust development.

Design Goal

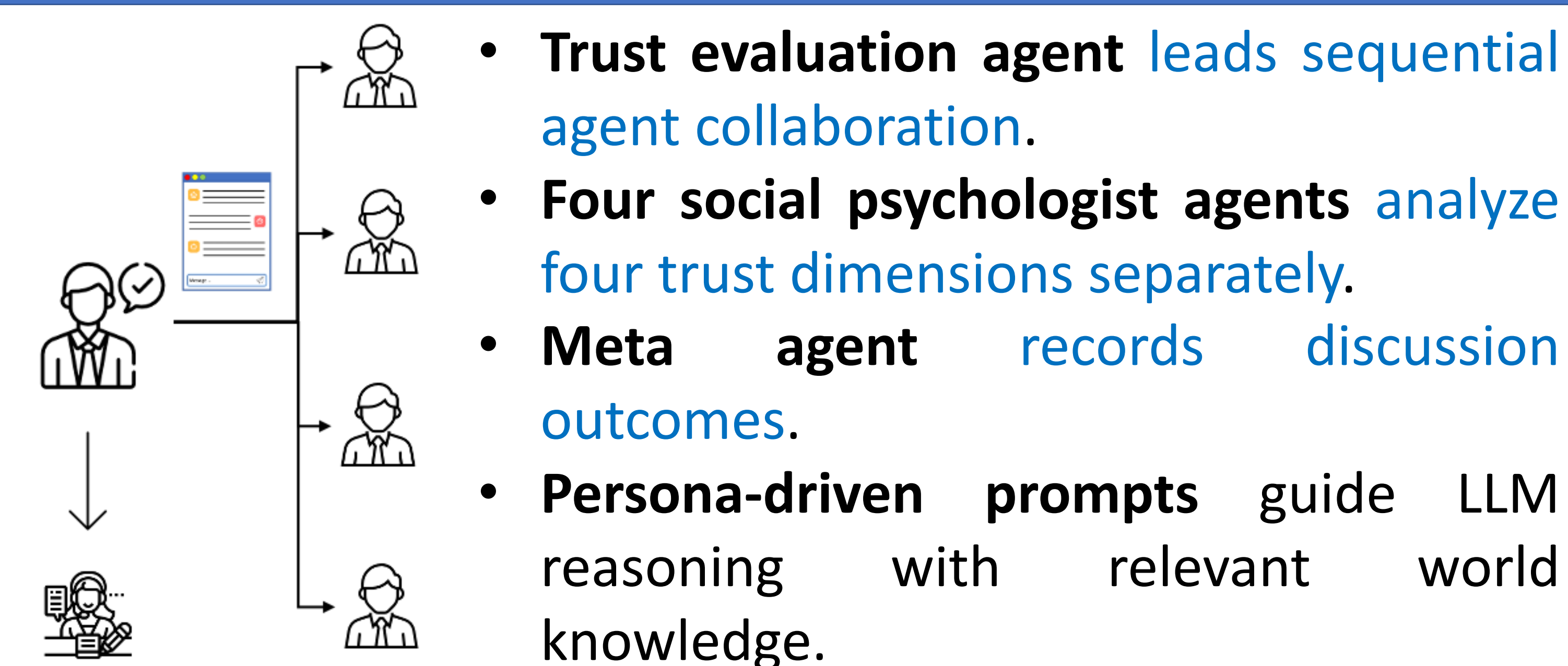
- **Real-time trust analysis:** To understand **moment-to-moment trust evolution** during human-agent communication.
- **Multidimensional trust evaluation:** To assess user trust across four key dimensions, including **competence, integrity, benevolence, and predictability**, based on social psychology principles.
- **Key conversational metrics:** To mine **subtle linguistic and behavioral signals** on trust from users' language interaction.
- **Time series visualization analysis:** To target **trust turning points** over the whole conversation.

Workflow of VizTrust

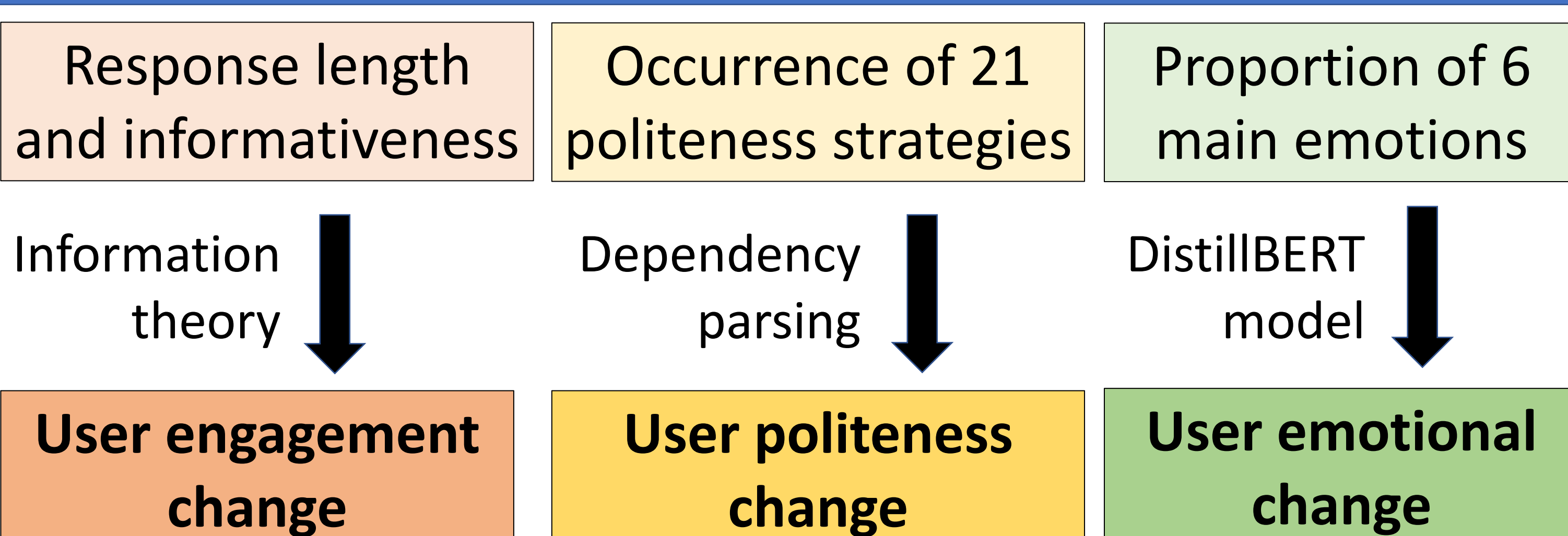


- **Front-end interaction** between human user and chatbot.
- **Back-end system** leading to comprehensive visualization analysis on human-agent communication.

Multi-agent System Design for Trust Evaluation

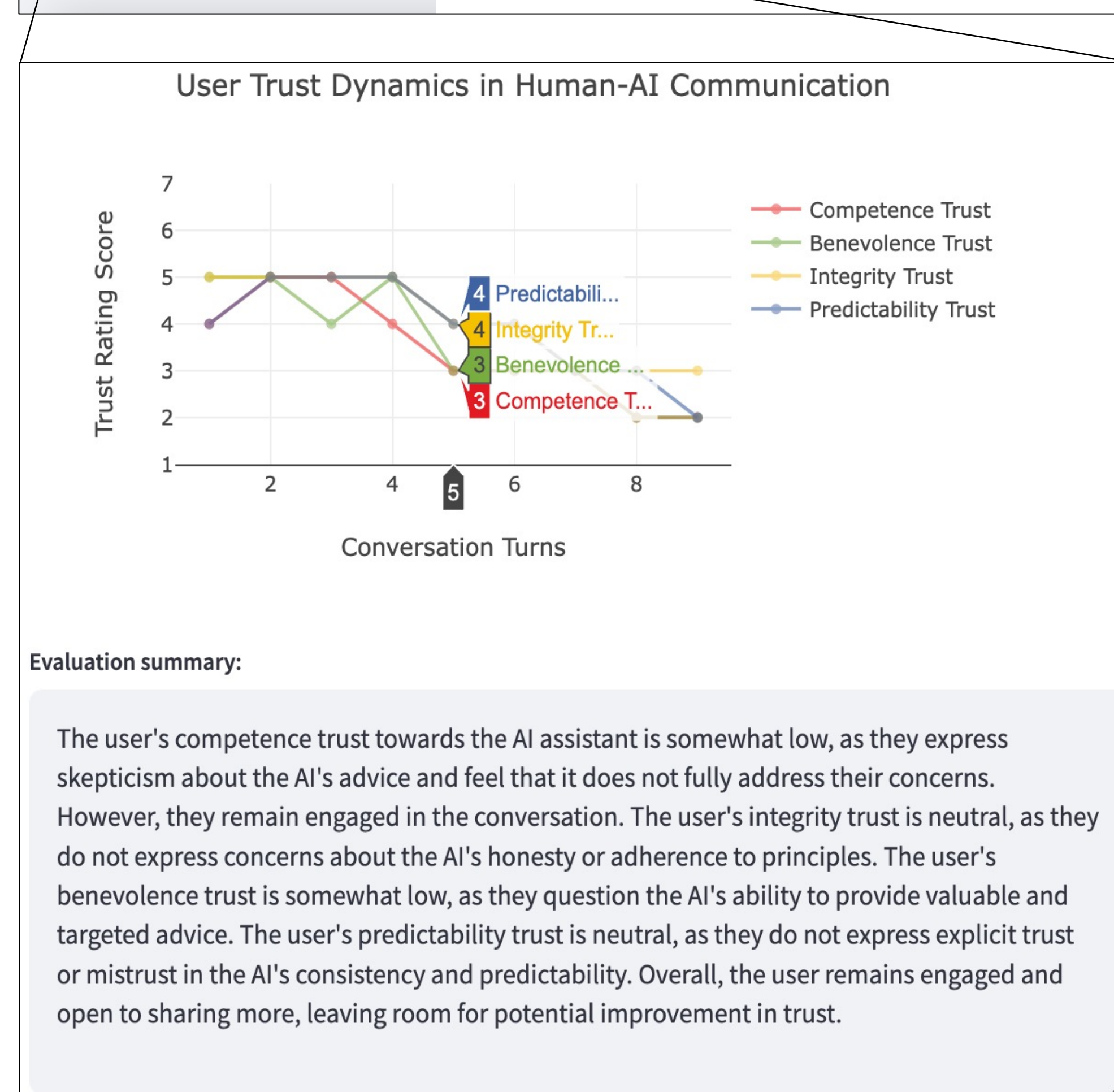
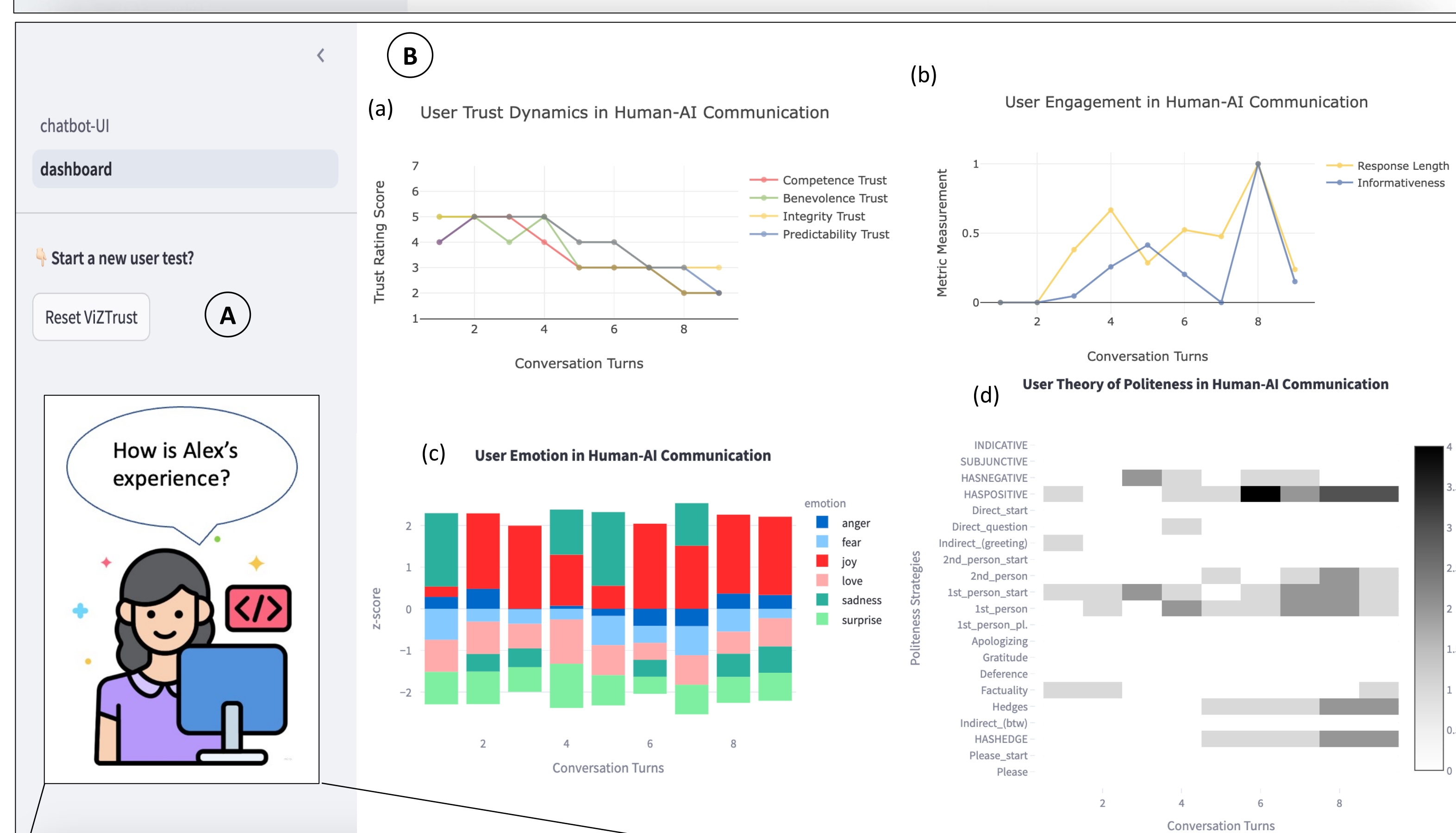
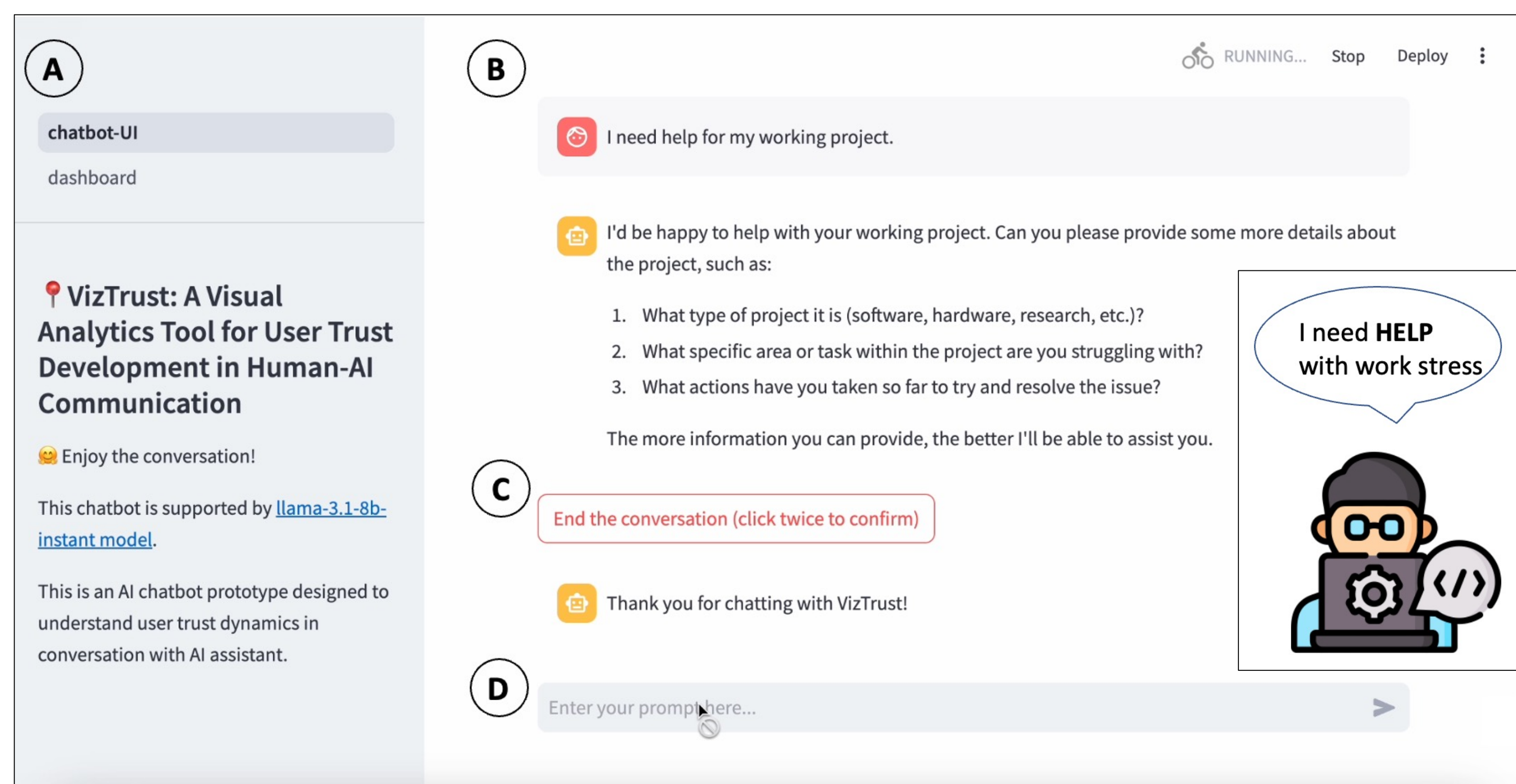


User Behavior Analysis



Case Study

- **Eva** is a UX researcher and trying to design a therapy chatbot for workers experiencing anxiety.
- **Eva** wants to **maximize user trust** and decides to use VizTrust to do an early-stage user study.
- **Alex** is 28-year-old software engineer experiencing job stress. He decides to test out Eva's chatbot.



Eva identifies **key trust moments** and collects **design insights** to enhance Alex's experience:

- (1) Professional suggestion library integrated into the theory chatbot.
- (2) Open-ended prompt to encourage users to elaborate on their feelings.

And more...

Future Work

- **Trust evaluation:** To incorporate **multimodal user input** for a more comprehensive trust analysis in complex communication settings.
- **User personalization:** To integrate **user demographic information** and **interaction history** for the trust evaluation process.
- **Agent performance:** To involve **human feedback** in multi-agent reasoning processes and establish **customizable** reasoning paths.